

1.0 Objective

The objective of this lab is to understand how thermal systems which involve thermal masses are used to transfer heat and energy in thermal systems. Over the course of this lab a thermal system simulated in Simscape will be used to simulate a thermal transient system response within a thermal mass.

2.0 Background Theory

Thermal systems rely on several equations that govern how heat and thermal energy is transferred through objects of different properties such as thermal conductivity and resistance.

Thermal Resistance

Thermal resistance is defined as the ability of a material to resist conductive heat transfer and is governed by equation (1). In this case L represents the thickness of the material, k is a constant of thermal conductivity, and A is the area of which the thermal flux travels through.

$$R = \frac{L}{kA} \quad (1)$$

Thermal Capacitance

Thermal capacitance can be modeled by equation (2) and represents the amount of energy stored in a given mass for every degree of temperature K that it is raised. In this equation m represents the mass of the object, and c_p represents the material's specific heat.

$$C = mc_p \quad (2)$$

Transient Heat Transfer

Due to the nature of transient processes, Transient Heat Transfer must be modeled as a differential equation. As shown in equation (3) the $C \frac{dT}{dt}$ term, also known as heat flow, can be described simply as the change in heat through an object divided by its Thermal Resistance.

$$C \frac{dT}{dt} = \frac{T_0 - T_i}{R} \quad (3)$$

Thermal Time Constant

The Thermal time Constant can be easily calculated by multiplying the Thermal Resistance R , and Thermal Capacitance C as shown in equation (4).

$$\tau = RC \quad (4)$$

3.0 Model Description

Different components within Simscape were used to model and measure the transient thermal response generated from the two thermal masses. The system being modeled includes a room and a wall acting as a barrier between the room and outside as shown in Fig.1. The outside of the wall acts as a temperature source, which brings heat into the room via convection of the air and conduction with the wall mass to heat the mass within the room. This is modeled using several Simscape blocks with a thermal reference and a scope to measure the temperature of the wall and room.

Simscape Components

- Temperature Source - generates heat
- Thermal Mass - acts as a thermal capacitor, storing heat
- Conductive Heat Transfer - transfers thermal energy via direct contact
- Temperature Sensor - models sensor instrumentation such as thermometers
- Thermal Reference - creates a reference temperature used to find actual temperature
- PS-Simulink Converter - converts Simscape signal to Simulink signal
- Scope - generates a graph based on the output temperature signal over time

4.0 Simscape Diagram

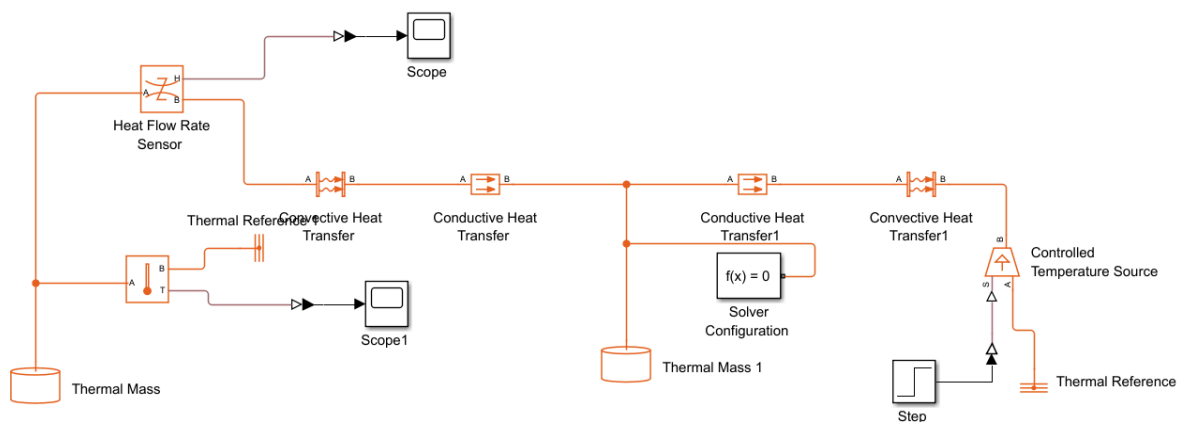


Fig.1 Thermal Simscape Diagram

5.0 Parameters and Setup

Initial Room Temp	$T_i = 300 \text{ K}$	Wall Thickness	0.3 m
Initial Wall Temp	$T_m = 300 \text{ K}$	Wall Conductivity	0.005 W/m/K

Thermal Capacities	C1=1000 J/K, C2=100 J/K	Outside Temperature	250 K
Wall Area	160 m ²	Simulation Time	600 seconds

Table 1. System Parameter Values

6.0 Results

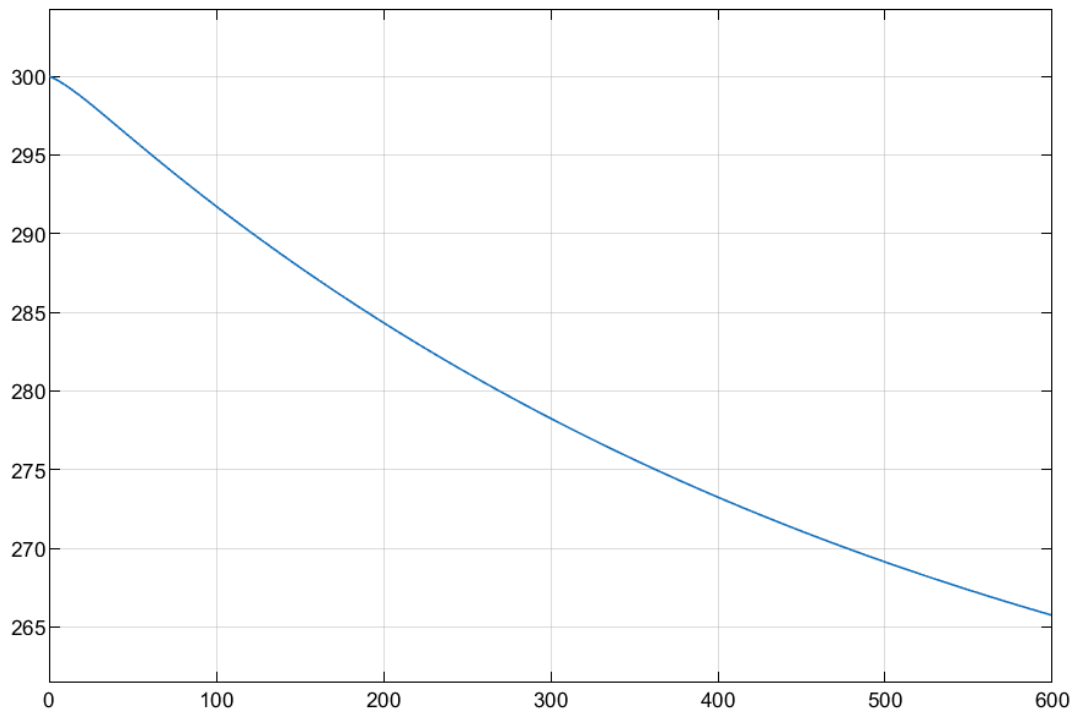


Fig. 2 Heat Flow Rate Over Time

Fig. 2 shows that over time the amount of heat flowing through the wall actually decreases very steeply as the temperature between indoors and outdoors begins to converge to the same value.

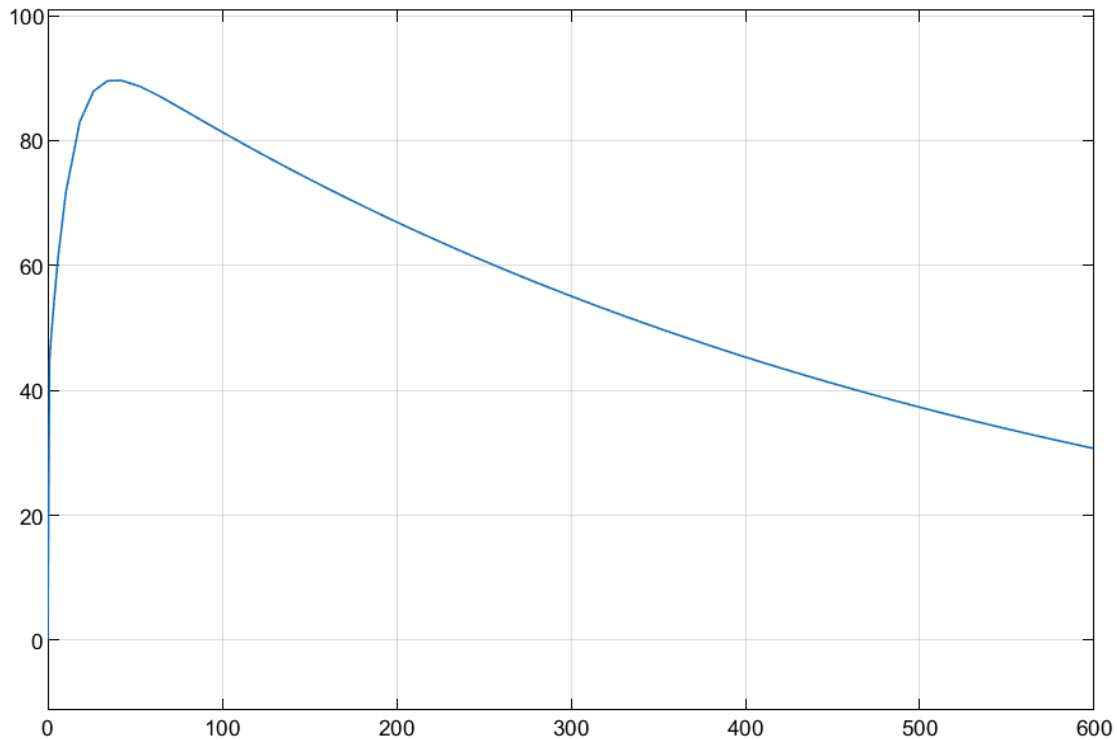


Fig. 3 Room Inside Temperature Over Time

Fig.3 Shows a steady decline of temperature inside of the room over time. This makes sense because heat is leaving the room to the cooler outside air. You can also see that overtime the change in temperature begins to plateau as the temperature of indoors and outdoors becomes similar.

6.0 Discussion

The main reason why the temperature of the room decreased over time is due to the initial temperature differential of the outside and inside air. The thermal resistance of the wall slows down the overall temperature change process. However, as the temperature drops over time, so does the thermal capacitance and heat flow. Thus, as time goes on and the temperature decreases, less and less heat will flow through the wall through conductivity.

7.0 Conclusion

Overall, this lab was a good introduction to thermal systems and initialization functions. Whereas the Simscape model itself helps the lab user to better understand the necessary components of a Thermal system and how that system represents the fundamental laws that dictate a real setting between objects of different temperature, thermal resistance, and thermal capacitance.